Google Cloud H2S

AI Builder Cup

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Veriqo

Know what to check before you buy.

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Problem Statement

Used-car buyers can't quickly tell what a car's photos actually show, what its registration record says, or what still needs checking — so they overpay or buy hidden problems. Veriqo gives a fast, evidence-based pre-purchase report that is honest about what it could not verify.

Brief about the idea

Veriqo turns a used-car buyer's photos, the registration number and the seller's listing into one evidence-tagged report — measured median 9.8 seconds from upload to report on the live deployment.

- **Gemini 2.5 Flash reads every photo:** visible damage, rust, tyre wear and interior damage — each finding linked to the photo it came from.
- **A registration lookup adds record facts** (insurance, loan, challans, blacklist). In this prototype it is SIMULATED with synthetic test plates; any other plate is shown as “not verified”.
- **Deterministic rules** build a transparent score, flag a listing that doesn't match the record, and flag missing photo angles.
- **Every finding is labelled** Observed · Record · Inferred · Not checked — and every report ends with a prioritised “what to check before paying” list.

Veriqo is a red-flag triage. It never certifies a car as safe — accident history and hidden mechanical condition are always listed as Not checked, so a clean-looking car never gets an “all clear”.

Opportunities

- **How different is it?** Marketplaces inspect only the cars they sell; independent inspections are paid, physical and take a day. Veriqo is an instant, self-serve first check for any car — including a private listing.
- **How does it solve the problem?** Before any money or travel, the buyer sees what the photos show, what the record says, what doesn't match, and exactly what to inspect.
- **USP — evidence you can trace.** Every finding says where it came from and what it's based on; the score is the sum of published rules; anything simulated or unverified is labelled as such.

The four provenance labels used in every report:

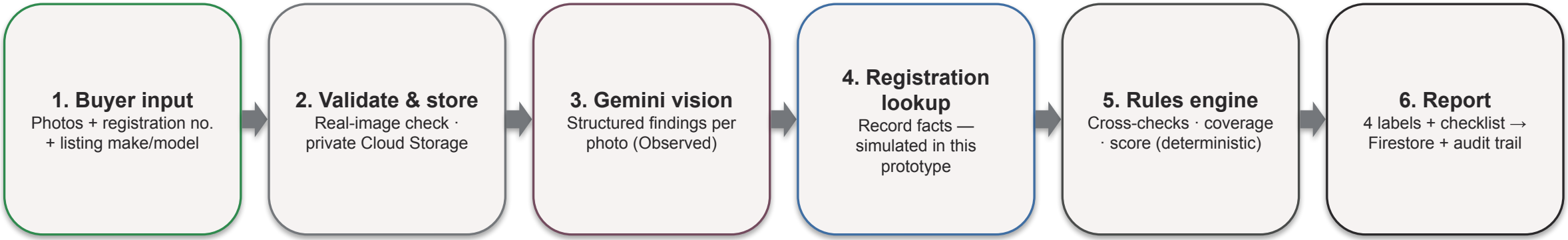
Observed | Record | Inferred | Not checked

List of features offered by the solution

- **Photo analysis with evidence.** Each finding shows the photo it came from, with Gemini's confidence.
- **Registration check.** Insurance, loan, challans, blacklist — simulated in this prototype and labelled so.
- **Listing vs record check.** Flags when the seller's make/model differs from the record.
- **Missing-angle detection.** Says which views (rear, sides, tyres, interior) were never shown.
- **Transparent score.** Sum of published rules; photos alone can't max it out (capped at 60).
- **Prioritised checklist.** “What to check before paying”, highest-impact first.
- **Saved, shareable report.** Stored in Firestore; the link reopens it after a refresh.
- **Safe failure handling.** Rejects non-images and >12 MB files; clear messages if the model or network fails.

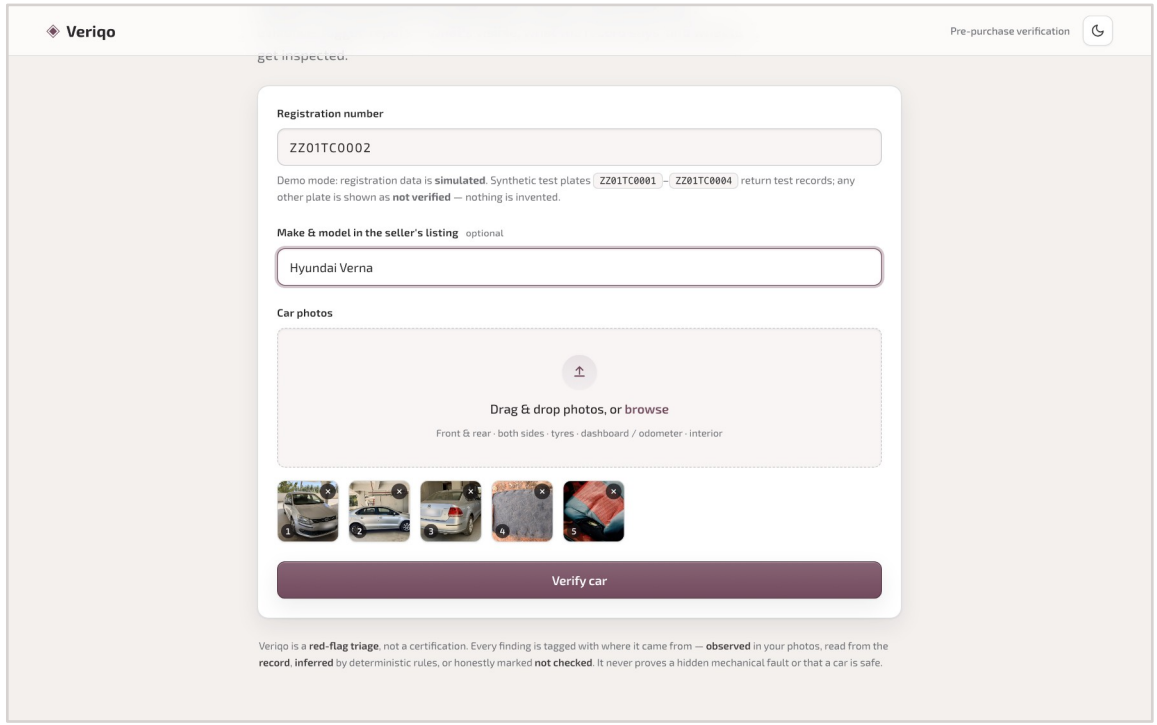
Honesty built in: simulated registration data is labelled “SIMULATED — NOT GOVERNMENT VERIFIED”; an unknown plate is shown as “not verified” — never filled with invented data.

Process flow diagram

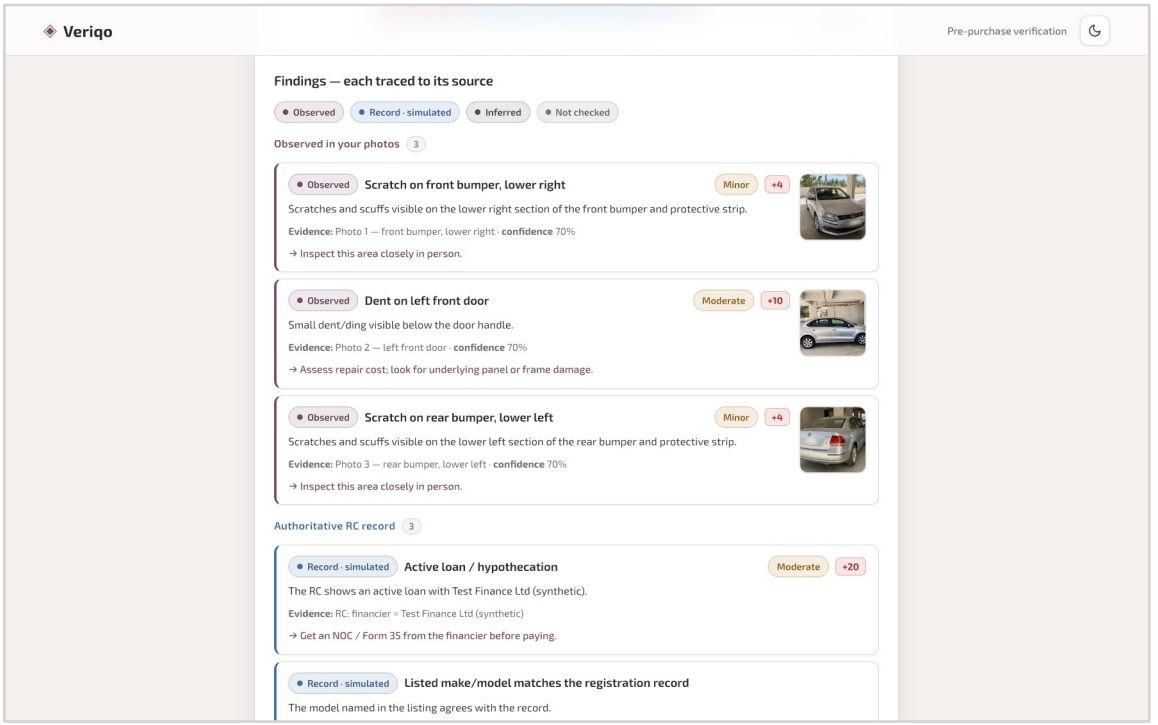


One buyer action in, one traceable report out. If Gemini fails, nothing is saved and the buyer sees a clear retry message.

Wireframes / Mock diagrams of the proposed solution



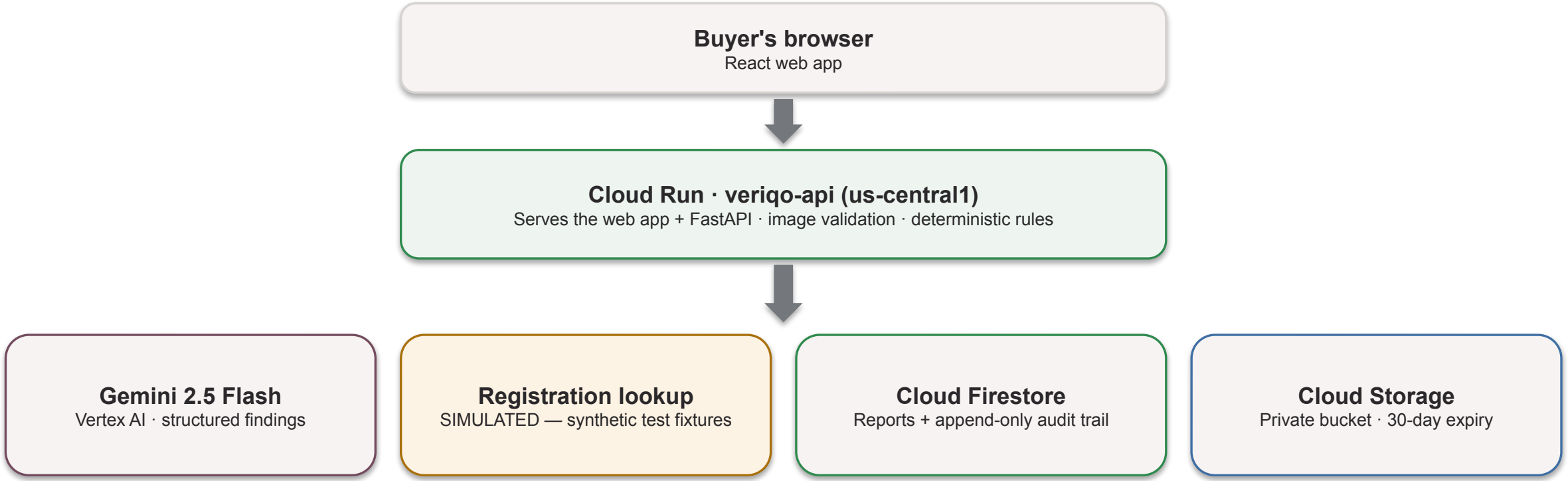
Upload: plate, listing, numbered photos



Report: findings labelled by source, with photo evidence

Actual screens captured from the deployed app (veriqo-api-920194228751.us-central1.run.app) — not mockups.

Architecture diagram of the proposed solution



Gemini perceives; Cloud Run decides with fixed rules; Firestore keeps the auditable record. Not used in this prototype: Firebase Hosting and Firebase Authentication (Cloud Run serves the app; there is no sign-in).

Technologies to be used in the solution

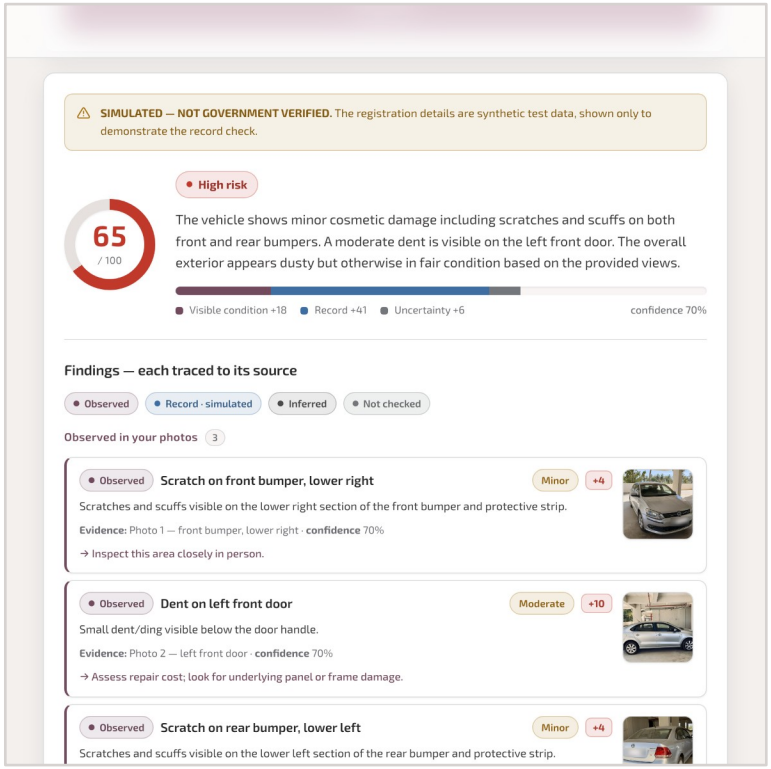
- **Gemini 2.5 Flash on Vertex AI —** multimodal photo analysis with a typed JSON response schema (temperature 0).
- **Cloud Run —** one service serving the React app and the FastAPI backend; 1 warm instance.
- **Cloud Firestore (Firebase) —** saved reports and an append-only audit trail.
- **Cloud Storage —** uploaded photos in a private bucket, deleted automatically after 30 days.
- **Cloud Build + Artifact Registry —** builds and stores the container on every deploy.
- **React + Vite + TypeScript · FastAPI + Pydantic (Python) —** app and API.
- **pytest + Playwright —** API tests and browser tests run against the live deployment.
- **Registration data —** synthetic fixtures today; built to plug in a licensed provider (Surepass / Signzy class).

Estimated implementation cost (optional)

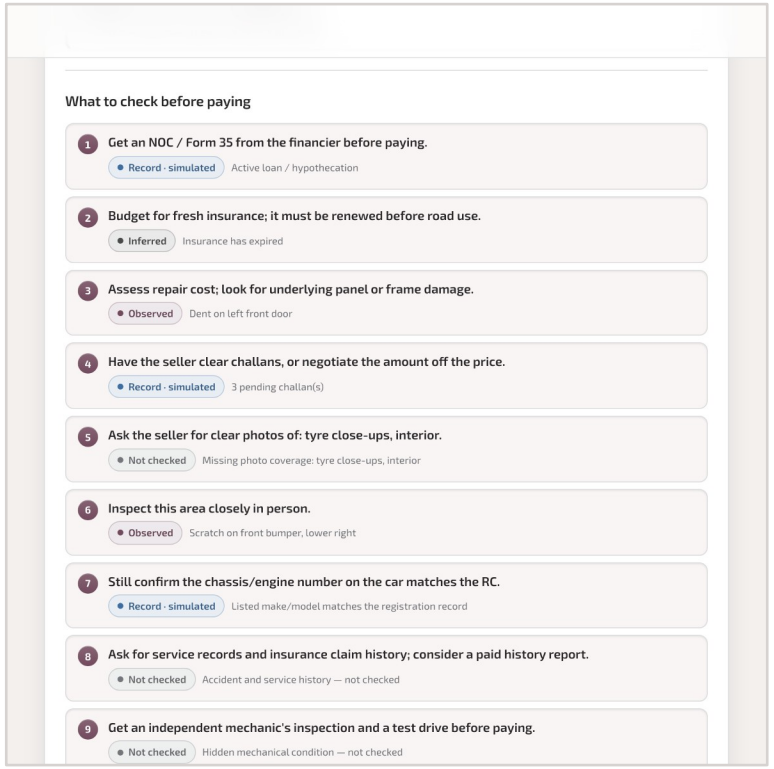
Indicative, prototype scale. These are estimates, not measured bills.

Component	Driver	Indicative cost
Gemini 2.5 Flash	≈5 photos per report	Fractions of ₹ per report
Cloud Run	1 always-warm instance (avoids cold starts)	Small fixed monthly cost
Firestore + Cloud Storage	Prototype volumes; photos auto-deleted at 30 days	Within free tier
Registration provider (when live)	Per-lookup fee	Likely the largest per-report cost

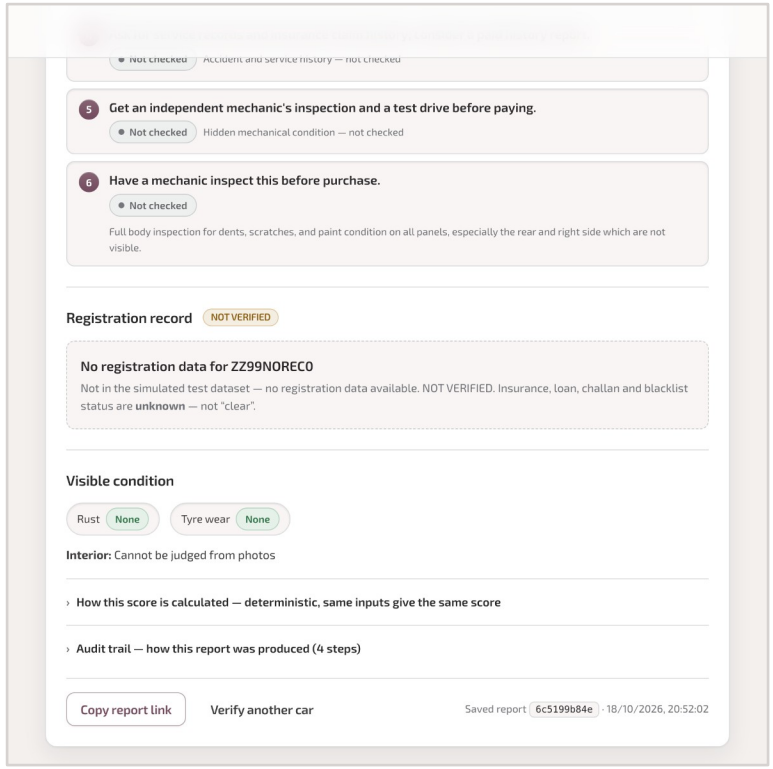
Snapshots of the prototype



Score, breakdown, simulated label



Prioritised buyer checklist



Unknown plate → not verified

Captured from the live deployment with real Gemini output. Test photos: openly licensed Wikimedia Commons images (credits in the repo). Registration data is synthetic and labelled as simulated.

Prototype Performance report/Benchmarking

What we measured	Result
API tests (pytest, 21 cases incl. invalid files, model failures, persistence)	21 / 21 pass
Scoring-rule evaluation (11 labelled cases)	67 / 67 checks pass
Browser tests on the live deployment (15 scenarios)	15 / 15 pass (first run: 14 / 15)
Upload → report time, 13 live runs	median 9.8 s · range 6.8–17.2 s

Model review (checked by hand, before → after prompt and rule fixes)

- **Fixed:** one key scratch scored 100/100 → now 26 (one moderate item); sharp close-ups were marked low-confidence → now 0.7–1.0; a blurry photo claimed views → now claims none; repeat runs disagreed → now identical.
- **Still open:** headlight severity varies between photo sets; a dent photo still yields 6 items. No precision/recall is claimed — 11 photos are not a labelled benchmark.

Sources: docs/05-TEST-RESULTS.md · docs/05b-MODEL-REVIEW.md

Additional Details/Future Development

- **Live registration data.** Connect a licensed provider so records become “Record verified”, not simulated.
- **A real accuracy benchmark.** ~100 labelled photo sets checked by mechanics → per-defect precision/recall.
- **Accounts.** Firebase Authentication so buyers can keep and share their reports.
- **History signals.** Partner data for accident, service and odometer history.
- **Local languages.** Reports in Hindi and regional languages with Gemini.
- **Same engine, new listings.** Verify rental-flat listings with the same evidence-tagged approach.

Provide links to your

- **GitHub Public Repository** https://github.com/radhikaaudi/Veriqo
- **Demo Video Link (3 Minutes)** https://youtu.be/zZbVIwDWArE
- **Final Product Link** https://veriqo-api-920194228751.us-central1.run.app

The live app is kept warm (1 always-on Cloud Run instance) for the evaluation period.

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Thank you

